Lab#05

Manipulation of type casting in Java

Objective

To get familiarity with the type casting in java.

Lab Goals

To provide working insight into the usage of type casting in java programming:

a. Demonstrate the implementation of type casting.

b. Explore implicit and explicit type casting in Java

Introduction to Java Type Casting

Type casting is when you assign a value of one primitive data type to another type. In Java, there are two types of casting:

- Widening Casting/ Implicit casting (automatically): Converting a smaller type to a larger type size.

byte -> short -> char -> int -> long -> float -> double

- Narrowing Casting/ Explicit casting (manually): Converting a larger type to a smaller size type.

double -> float -> long -> int -> char -> short -> byte

Widening Casting/ Implicit casting

Widening casting is done automatically when passing a smaller size type to a larger size type:

Example#1

```
public class Main {
    public static void main(String[] args){
        int myInt =9;
        double myDouble = myInt; // Automatic casting: int to double
        System.out.println(myInt); // Output 9
        System.out.println(myDouble); // Output 9.0
    }
}
```

Output

```
9
9.0
```

Exercise

A. Why does Java allow implicit casting from **int** to **double** but requires explicit casting from **double** to **int**? Explain reason in maximum 3 lines with example code justification.

1. **int** is smaller (32-bit) and **double** is larger (64-bit), so **int** → **double** never loses data — Java does it automatically.
2. But **double** → **int** cuts off decimals (e.g. *9.99* becomes *9*), so data is lost — Java forces explicit cast **(int)** to warn you.
3. This rule is called **Widening** (automatic) vs **Narrowing** (manual) casting.

Example Code:

```
public class Lab5 {
    // Example Code
    int a = 10;
    double b = a;          // Implicit (widening) — works automatically → 10.0
    double x = 9.99;
    int y = (int) x;       // Explicit (narrowing) — must cast manually → 9
}
```

B. Create a program that demonstrates implicit casting by:

- Assigning a **byte** value to an **int** variable
- Assigning an **int** value to a **long** variable
- Assigning a **float** value to a **double** variable

Print all values before and after conversion.

Program:

```
public class Lab5 {

    Run | Debug
    public static void main(String[] args) {

        // 1. Assigning a byte value to an int variable
        byte byteValue = 10;
        int intValue = byteValue;   // implicit casting: byte -> int
        System.out.println(x: "--- byte to int ---");
        System.out.println("Before (byte) : " + byteValue);
        System.out.println("After  (int)  : " + intValue);

        System.out.println();

        // 2. Assigning an int value to a long variable
        int intNum = 5000;
        long longNum = intNum;      // implicit casting: int -> long
        System.out.println(x: "--- int to long ---");
        System.out.println("Before (int)  : " + intNum);
        System.out.println("After  (long) : " + longNum);

        System.out.println();

        // 3. Assigning a float value to a double variable
        float floatValue = 3.14f;
        double doubleValue = floatValue;  // implicit casting: float -> double
        System.out.println(x: "--- float to double ---");
        System.out.println("Before (float)  : " + floatValue);
        System.out.println("After  (double) : " + doubleValue);
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 5> javac Lab5.java
PS C:\Users\Ansharah\Desktop\Java\Lab 5> java Lab5
  --- byte to int ---
  Before (byte) : 10
  After  (int)  : 10

  --- int to long ---
  Before (int)  : 5000
  After  (long) : 5000

  --- float to double ---
  Before (float)  : 3.14
  After  (double) : 3.140000104904175
PS C:\Users\Ansharah\Desktop\Java\Lab 5>
```

C. Create a program that demonstrates explicit casting by
- Converting a **double** to an **int**
- Converting a **long** to a **short**
- Converting a **float** to a **byte**

Print all values before and after conversion

Program:

```java
public class Lab5 {

    Run | Debug
    public static void main(String[] args) {

        // 1. Converting double to int
        double doubleValue = 9.99;
        int intValue = (int) doubleValue;   // explicit casting: double -> int
        System.out.println(x: "--- double to int ---");
        System.out.println("Before (double) : " + doubleValue);
        System.out.println("After  (int)    : " + intValue);

        System.out.println();

        // 2. Converting long to short
        long longValue = 100000L;
        short shortValue = (short) longValue;  // explicit casting: long -> short
        System.out.println(x: "--- long to short ---");
        System.out.println("Before (long)  : " + longValue);
        System.out.println("After  (short) : " + shortValue);

        System.out.println();

        // 3. Converting float to byte
        float floatValue = 130.75f;
        byte byteValue = (byte) floatValue;   // explicit casting: float -> byte
        System.out.println(x: "--- float to byte ---");
        System.out.println("Before (float) : " + floatValue);
        System.out.println("After  (byte)  : " + byteValue);
    }
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 5> java Lab5
 --- double to int ---
 Before (double) : 9.99
 After  (int)    : 9

 --- long to short ---
 Before (long)  : 100000
 After  (short) : -31072

 --- float to byte ---
 Before (float) : 130.75
 After  (byte)  : -126
PS C:\Users\Ansharah\Desktop\Java\Lab 5>
```

D. Write a program that calculates the average of three integers and stores the result in an integer variable using explicit casting. Then recalculate storing the result as a double. Compare both results by finding the difference between both the results.

Program:

```java
public class Lab5 {

    Run | Debug
    public static void main(String[] args) {

        // Three integers
        int a = 10, b = 20, c = 30;

        System.out.println("Three numbers: a=" + a + ", b=" + b + ", c=" + c);
        System.out.println(x: "========================================");

        // Average stored as int (using explicit casting)
        int intAverage = (int) ((a + b + c) / 3.0);
        System.out.println("Average as int    : " + intAverage);

        // Average stored as double
        double doubleAverage = (a + b + c) / 3.0;
        System.out.println("Average as double : " + doubleAverage);

        System.out.println(x: "========================================");

        // Difference between both results
        double difference = doubleAverage - intAverage;
        System.out.println("Difference        : " + doubleAverage + " - " + intAverage + " = " + difference);
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 5> Java Lab5
Three numbers: a=10, b=20, c=30
========================================
Average as int    : 20
Average as double : 20.0
========================================
Difference        : 20.0 - 20 = 0.0
PS C:\Users\Ansharah\Desktop\Java\Lab 5>
```

E. Write a program for a shopping cart that:

a. Takes item prices as doubles from the user input.
b. Calculates total as an integer (using explicit casting)
c. Calculates tax as a double
d. Shows how casting affects the final amount

Hint:

- Calculate total with explicit casting to **int**
- Calculate tax (8%) as **double**
- Calculate final amount

Expected Output:

Enter price of item 1: 19.99
Enter price of item 2: 5.50
Enter price of item 3: 12.49

Subtotal (as double): 37.98
Subtotal (cast to int): 37
Tax (8%): 2.96
Final amount: 39.96

Program:

```java
public class Lab5 {

    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Take item prices as doubles from user
        System.out.print(s: "Enter price of item 1: ");
        double item1 = sc.nextDouble();

        System.out.print(s: "Enter price of item 2: ");
        double item2 = sc.nextDouble();

        System.out.print(s: "Enter price of item 3: ");
        double item3 = sc.nextDouble();

        // Subtotal as double
        double subtotalDouble = item1 + item2 + item3;

        // Subtotal cast to int (explicit casting)
        int subtotalInt = (int) subtotalDouble;

        // Tax 8% calculated on int subtotal
        double tax = subtotalInt * 8.0 / 100;

        // Final amount
        double finalAmount = subtotalInt + tax;

        // Print results
        System.out.println();
        System.out.println("Subtotal (as double): " + subtotalDouble);
        System.out.println("Subtotal (cast to int): " + subtotalInt);
        System.out.println("Tax (8%): " + tax);
        System.out.println("Final amount: " + finalAmount);

        sc.close();
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 5> javac Lab5.java
PS C:\Users\Ansharah\Desktop\Java\Lab 5> java Lab5
Enter price of item 1: 99
Enter price of item 2: 890
Enter price of item 3: 792

Subtotal (as double): 1781.0
Subtotal (cast to int): 1781
Tax (8%): 142.48
Final amount: 1923.48
PS C:\Users\Ansharah\Desktop\Java\Lab 5>
```